Algorithmic Optimisation or Relational Pedagogy? AI, Human Connection and the Formation of Self in the Learner-Teacher Relationship

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Abstract

This paper examines what a developing sense of self depends on when artificial intelligence (AI) is positioned within the learner-teacher relationship. AI-in-education research has emphasised personalisation, efficiency and human oversight, but has rarely engaged the developmental-psychology evidence that teaching is constituted through relational security, co-regulation and mentalisation rather than merely accompanied by them. Synthesising work across human-centred AI, developmental psychology and participation theory, the paper distinguishes a cognitive-support layer AI can occupy from a relational one it cannot; inference and synthesis of affective state approximate the outward form of attunement without the mind-reading and responsiveness that relational security depends on. Evidence that adults track identity continuity in AI companions and respond to its disruption with reactions characteristic of relationship loss suggests these self-formation stakes are not speculative when a system is placed where a relationship should be. The paper is theoretical rather than empirical, and its conclusions are provisional; it argues that AI may support thinking about teaching without constituting the relational work through which a learner's sense of self is formed, cautioning against both technological determinism and reflexive dismissal of AI's support functions.

1 Introduction

Artificial intelligence (AI) in education is increasingly framed through the language of personalisation, efficiency, prediction and assessment (Alfredo *et al*., 2024), positioning learning as something to be optimised: inputs adjusted, pathways individualised, outcomes measured. A substantial body of developmental psychology positions teaching differently, not as a transaction between a system and a learner but as a relationship, constituted through attunement, co-regulation and mutual interpretation (Bowlby, 1969; Bergin and Bergin, 2009; Kostøl and Mänty, 2024). These are not simply different emphases within one picture; they rest on different assumptions about what learning fundamentally is, and reconciling them is not straightforward.

This is also a question about self-formation. Attachment and mentalisation research treats being read and responded to by another mind as developmentally significant for a child's sense of self, not only for behaviour, attainment or social adjustment (Bowlby, 1969; Cecere *et al.*, 2025). A framework that does not address this addresses only part of what the relationship is for.

This paper argues that education cannot be adequately understood, or improved, through data-driven optimisation alone, because teaching is constituted through situated human relationships that current AI systems do not model. This is not a claim that AI has no place in education: human-centred AI (HCAI) research itself insists on trust, human oversight and meaningful human control (Alfredo *et al*., 2024), and an established critical literature already addresses AI in education on grounds of

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datafication, platformisation and the political economy of provision (Selwyn, 2019; Williamson, 2024). The claim here is narrower: the relational work of teaching, evidenced in research on learner-teacher mentalisation and closeness (Cecere *et al*., 2025) and practitioner co-regulation of children's emotional expression (Burgess, 2023), has not been brought to bear on the frameworks currently shaping AI in education. UNESCO's teacher-AI-student competency framework (Miao and Cukurova, 2024), for instance, acknowledges that AI reconfigures the traditional teaching relationship into a three-way dynamic, but does not engage attachment- or mentalisation-based theory in doing so, a gap this paper takes as its starting point.

The argument is developed with reference to Special Educational Needs and Disability (SEND) contexts, where relational dependency is rendered especially visible (Shildrick and Price, 2006; Goodley and Runswick-Cole, 2013), though the paper's scope is not confined to SEND or to any single phase of education. The evidence base for attachment-informed practice remains strongest in early childhood (Sabol and Pianta, 2012; Verschueren, 2015); its extension across the wider learner-teacher relationship is itself part of what this paper examines, rather than assumed in advance.

2 Literature Review and Theoretical Framework

2.1 AI in education and the question of trust

HCAI research in education has increasingly emphasised trust, human oversight, and the preservation of meaningful human control over automated systems (Alfredo *et al*., 2024). Alfredo *et al.* map educational AI across two dimensions, human control and computer automation, reporting mixed findings on stakeholder trust and uneven attention to safety, reliability and trustworthiness across the field. The framework treats control and automation not as a trade-off but as independent dimensions, on which well-designed systems can rank highly on both (Shneiderman, 2022).

Shneiderman frames the underlying question as whether people belong in the same category as computers or are distinct from them, setting the Stanford AI-100 report's contention that human and machine intelligence differ in scale, speed, autonomy and generality rather than in kind against the position, which he shares, that the difference is vast (Shneiderman, 2022, p. 25); his grounds are not metaphysical, since human capabilities, shaped by lengthy physical and cultural evolution, warrant appreciation on their own terms rather than by appeal to a soul or spirit. This paper proceeds on the same footing, but the framework's limits lie in its own construction: both dimensions describe a relation between a user and a system, and a relation between two people, which a system might mediate, thin or displace, is not a quantity those axes can express. This is a disciplinary boundary, not an oversight: human-computer interaction asks what a system does for its user, while what teaching demands between people is a question developmental and educational psychology answers differently.

This paper therefore introduces its own distinction, developed through the sections that follow. A cognitive-support layer assists the work of teaching – reflection, planning, differentiation – while pedagogical and relational judgement remains with the practitioner. A relational substitute occupies the place of the person the learner is in relationship with – a difference not of sophistication but of position.

2.2 Relational foundations: attachment, attunement and co-regulation

Bowlby's (1969) attachment theory positions relational security, established through consistent, responsive caregiving, as foundational to a child's capacity to explore, learn and regulate emotion. Although Bowlby's original formulation concerns parent-infant attachment, subsequent research extends this framework into classroom relationships: a review of the learner-teacher relationship literature shows that relationships carrying attachment-relevant qualities – security, closeness, low conflict – predict children's engagement, achievement, emotion regulation and social competence at school, while cautioning that most such relationships are better described as attachment-like than as full attachment bonds (Bergin and Bergin, 2009). If attuned relationships with a teacher, not merely a caregiver, meaningfully predict learning outcomes, then any account of AI's role in education must contend with whether, and how, AI can participate in that relational function.

This claim need not rest on a single source. Verschueren's (2015) review of teacher-child relationship research in middle childhood states the position more precisely: the teacher-child relationship does

not constitute a full attachment bond in Ainsworth's (1989, cited in Verschueren, 2015) sense of an enduring tie to a partner interchangeable with no other, since teachers are typically neither exclusive nor durable, but positions the teacher as a temporary or ad-hoc attachment figure providing secure base and, decreasingly through middle childhood, safe haven functions. As children move from early into middle childhood, the attachment system's goal shifts from the caregiver's physical proximity toward their availability (Kerns, 2008, cited in Verschueren, 2015), complicating any claim that a persistently accessible AI system might substitute for this function: availability is a felt sense of accessibility built cumulatively through a history of responsive interaction, not equivalent to uptime, and, as the Discussion argues, not the same as reliability.

Verschueren (2015) treats teacher sensitivity, rather than instructional input, as a central antecedent of relational quality. Buyse, Verschueren and Doumen's (2011, cited in Sabol and Pianta, 2012) finding that children with insecure maternal attachment developed less close teacher relationships only where observed teacher sensitivity was low suggests that what compensates for relational risk is responsive attunement, not guidance alone. These dimensions carry consequences beyond the dyad itself: observing 261 preschoolers, Prino *et al*. (2023) found that positive negotiation behaviours directed at teachers predicted children's social preference among peers, though the cross-sectional, single-country design limits generalisation.

Co-regulation research develops this further by examining how practitioners actively support children's emotional regulation through relational, embodied and often tacit means. Kostøl and Mänty (2024) demonstrate that teachers engage in ongoing, situated evaluative judgement when responding to children's emotional cues, drawing on relational sensitivity as much as formal training. Burgess (2023) similarly finds that educators' capacity to co-regulate children is itself supported by their own co-regulation with colleagues, positioning relational capacity as distributed across a practitioner community rather than a single, isolable skill. Neither study frames co-regulation as reducible to a defined procedure; both position it as an interpretive, relational practice, precisely the kind current AI systems are not designed to perform.

2.3 Mentalisation and relational closeness in inclusive practice

Cecere *et al*. (2025) examine mentalisation, the capacity to understand another's mental state as distinct from yet connected to one's own, and relational closeness between teachers and pupils in inclusive settings, finding both meaningfully associated with the quality of the learner-teacher relationship, closely related to attunement and co-regulation without being identical to them: mentalising well is arguably a precondition for attuned response.

This capacity has also been examined under the related construct of mind-mindedness: a caregiver's propensity to treat a child as an individual with a mind of their own, evidenced through appropriately attuned comments on the child's thoughts, desires and intentions. Colonnesi *et al*. (2017), in one of relatively few studies to examine mind-mindedness among professional rather than parental caregivers, observed 34 Dutch childcare practitioners in structured play with three-year-old children, finding that mind-related comments directed toward both children simultaneously were associated with practitioners' respect for children's autonomy and with children's attachment security, whereas comments directed toward one child individually were not. The authors read this as evidence that professional mind-mindedness is the capacity to tune in to more than one child at once, reflecting the group conditions under which educational relationships actually operate; this is a small, exploratory study in an atypical sample and its findings should not be over-generalised. It nonetheless raises a question personalisation-oriented accounts of AI do not currently address: if the relational mechanism most closely associated with security in group settings is non-dyadic, then technology designed around one-to-one adaptation is optimising for precisely the dyadic form of attention this evidence did not find associated with security.

2.4 Cognitive models and biopsychosocial accounts of learning

Engel's (1977) biopsychosocial model was formulated against biomedical reductionism: the assumption that disease can be understood through biological variables alone, with psychological and social dimensions peripheral. The parallel with educational technology is close: adaptive systems model knowledge states, error patterns and learning trajectories in detail, while affective and relational dimensions enter, if at all, as further variables to be encoded rather than as constitutive of what

is being modelled. On a biopsychosocial reading this is not an incomplete model of learning but a mistaken one, since the biological, psychological and social interact and constrain one another continuously. A system that models the cognitive well and the relational thinly has not modelled teaching less fully; it has modelled something else.

2.5 Participation, voice and the limits of recognition

Lundy's (2007) model of child participation, requiring space, voice, audience and influence for a child's views to constitute genuine participation under Article 12 of the United Nations Convention on the Rights of the Child (1989), offers a further lens: it asks not merely whether a learner's communication is registered, but whether it is recognised and responded to in a manner that carries weight. Cuevas-Parra's (2023) extension of this model adds attention to intersecting identities and enabling environments, cautioning that participation rights are unevenly realised even within human-mediated relationships, let alone AI-mediated ones. Neither addresses AI directly; their relevance here is diagnostic, establishing a standard against which AI-mediated interaction can be judged, without yet having been applied to it.

2.6 AI, participation and the reconfigured triad

Miao and Cukurova's (2024) AI competency framework for teachers explicitly acknowledges that AI reconfigures the traditional dyadic learner-teacher relationship into a three-way dynamic, a rare acknowledgement within AI-in-education policy literature, but does not engage attachment- or mentalisation-based theory in examining what this means relationally, nor draw on Lundy's (2007) participation model to ask whether an AI system can function as an 'audience' capable of exercising influence. Melo-López *et al.*'s (2025) systematic review similarly cautions that over-reliance on automation risks reducing direct human interaction, stated as a general principle rather than explored through developmental theory. The reconfiguration is not only relational: a third party to whom a learner may turn for knowledge introduces a question of epistemic authority alongside it, taken up in the Discussion.

A recent field-wide survey indicates how far this omission extends into current technical work. Chu *et al.* (2025) map educational large language model agents across teaching-assistance functions – classroom simulation, feedback generation, curriculum design – and student-support functions covering adaptive learning, knowledge tracing and error detection. Affective considerations appear, but instrumentally: agents maintain modular state components capturing a learner's cognitive, affective and psychological dimensions, formalised within a state-action framework in which learner traits become states against which instructional actions are selected (Chu *et al.*, 2025, para. 4.1). Attachment theory, mentalisation and the learner-teacher relationship literature are not engaged anywhere in the survey; it reports its field accurately, and the absence is evidence that the relational evidence base and the technical literature remain separate.

Regulation has moved on the same terrain. The EU Artificial Intelligence Act (Regulation (EU) 2024/1689) names education in two tiers: the Article 5(1)(f) prohibition on inferring emotions in education institutions has applied since 2 February 2025 and stands unamended, while the Annex III classification of systems evaluating learning outcomes as high-risk was deferred by sixteen months (Regulation (EU) 2026/1744), so the binding constraint on educational AI is currently the prohibition rather than the high-risk regime. Neither engages relational or developmental theory in doing so.

A partial, adjacent exception exists outside educational contexts. Wang *et al.'s* (2025) DinoCompanion presents an attachment-theory-informed companion robot for children, grounded explicitly in Bowlby's constructs of secure base and safe haven, demonstrating that attachment constructs can be operationalised and benchmarked in system design, but within home and caregiving contexts, without engaging mentalisation theory, Lundy's participation model, or the pedagogical relationship this paper addresses. The convergence this paper argues for therefore remains, on the evidence reviewed, unaddressed.

3 Methodology/Approach

This paper is a conceptual, practice-informed synthesis rather than a systematic review. It brings together HCAI research, developmental psychology, disability studies and participation theory, on

the biopsychosocial reasoning set out above: a question that is simultaneously technical, relational and rights-based cannot be adequately addressed from within any one of those literatures. The contribution is integrative rather than empirical, and its conclusions are theoretical.

The argument rests on a claim about absence: that attachment, mentalisation and co-regulation research has not been brought to bear on the design and evaluation of educational AI. Such claims are easily overstated, so this one is constructed deliberately, following Sandberg and Alvesson's (2011) distinction between gap-spotting, which identifies what is overlooked, under-researched or insufficiently applied and tends to reinforce existing theory since it leaves that theory's assumptions intact, and problematisation, which questions those assumptions directly. Stated as gap-spotting, this paper's claim would be that attachment and mentalisation theory are under-applied in AI-in-education research: true and uninteresting. Stated as problematisation, the claim is that an assumption runs through both the HCAI literature and the system design it evaluates: that relational qualities such as rapport, responsiveness and attunement are properties a system can be designed to possess, rather than properties of a relationship between people. It is that assumption, rather than any deficiency of coverage, that the paper questions.

The gap statement follows Wald, Harland and Daskon's (2024) four-part model: the context, the issue of concern, what is insufficiently known about it, and the contribution addressing it would make; their analysis of 124 articles in leading higher-education journals found that 27 per cent could not have claimed a worthwhile gap, and stating the four elements explicitly is intended to avoid that. Two further commitments follow. Gap claims are scoped to the searches actually conducted, expressed as what was not identified within particular terms, databases and date ranges rather than as an absence, and each was subjected to a second search seeking counter-examples before being retained. Empirical findings, theoretical argument, policy statement and practitioner opinion are treated as distinct categories and not substituted for one another; where a source that does not itself discuss AI is brought to bear on it, the inference is signalled as the authors' own.

SEND contexts are used as an illustrative lens, selected because relational dependency is rendered especially visible there, not because SEND learners are more relationally dependent than their peers. The lead author is an education practitioner; that background shapes the questions asked and is stated here as positionality, not as evidence.

4 Discussion

4.1 What AI can and cannot do relationally

The relational dimension of teaching is not being overlooked in AI development; it is being targeted. Selwyn (2019) documents developer ambitions for pedagogical agents that harvest emotionally charged data, employ biometric mood-detection techniques, and perform 'affect synthesis' in pursuit of agents that are 'life-like', 'natural and believable' (p. 61), some conceived as 'life-long' and 'life-wide' companions 'forging on-going relationships…just as people do' (p. 63).

Three distinct operations are involved, and each warrants scrutiny. The first is inference: reading a learner's internal state from external signals. Its reliability is contested from three directions: Shneiderman (2022) argues that machine reading of emotion oversimplifies and misses nuanced indicators; Alfredo *et al.* (2024) report teachers doubting systems to infer students' physiological states; and the EU AI Act (Regulation (EU) 2024/1689) cites serious concerns about the scientific basis of emotion inference, noting that emotional expression varies across cultures, situations and individuals, reasoning that places the practice in the prohibited tier for education institutions. The second is synthesis: the system performing affective expression of its own. Relational security is established through consistent, responsive caregiving from another person (Bowlby, 1969), and teacher mentalisation is empirically linked to relational closeness in inclusive settings (Cecere *et al*., 2025); synthesis produces the outward form of that responsiveness without the reading of another mind it depends on. The concern is not whether the performance is convincing, but what is substituted when it is.

Reliance is a third operation, following from the second. Fonagy and Allison (2014) position epistemic trust in another's knowledge as arising within attachment relationships, and as the condition under which a learner relaxes the epistemic vigilance that otherwise governs testimony (Sperber *et al*., 2010), a mechanism resting on being accurately represented in another mind. Synthesis therefore

presents the cues by which vigilance is ordinarily relaxed, toward a source that has not established the grounds for it, which bears on learners in particular since selective trust is itself a developing capacity (Harris *et al*., 2018). The source's unreliability is structural: Kalai and Vempala (2024) establish a statistical lower bound on false statements about arbitrary facts for calibrated pretrained models, while Kalai *et al*. (2025), in a preprint that has not been peer-reviewed, attribute the persistence of such errors to training and evaluation procedures that reward guessing over acknowledged uncertainty. The triad Miao and Cukurova (2024) describe is thus epistemic as well as relational, and the Department for Education's (DfE, 2026a) requirement that products remind learners AI cannot replace real human relationships is a disclaimer addressed to precisely those least equipped to act on it. None of these developmental sources addresses AI; the application here is the authors' own.

4.2 SEND as an illustrative lens

SEND contexts are used here to make something visible, not to locate it: relational dependency becomes more observable where communication is mediated, where regulation is supported, and where belonging cannot be assumed, not because learners with SEND are more relationally dependent than their peers.

Critical disability studies make this argument more strongly. Goodley and Runswick-Cole (2013) argue, on social psychoanalytic grounds, that fragmentation and lack are conditions of embodiment in general, and that non-normative bodies do not introduce these conditions but 'remind everyone of their own fragmented bodies that they have been forced to reject' (p. 10). Shildrick and Price (2006) make the same point about dependency: disabled people are not unique in relying on profound interconnectivity, but for the normative majority that reliance, 'inevitable as it is, may be covered over…' (para. 20). Neither addresses AI; the application here is the authors' own.

Theoretical visibility does not translate into measurable difference. Berchiatti *et al*. (2024), comparing 320 Italian students, found that the presence of SEN or learning disabilities did not significantly affect teachers' perceptions of closeness or conflict, though it did affect emotional and hyperactivity symptoms, attainment and peer acceptance, a finding that cautions against assuming what theory renders visible will also be empirically distinctive. Two readings of AI's contribution follow from this. Melo-López *et al.* (2025) find that AI supports accessibility and personalisation for learners with additional needs; Spaan, van Trigt and Schippers (2024), drawing on Dolmage's concept of the retrofit, argue that accessibility is not inclusion, since accommodation added after the fact singles out the body that must request access. Whether AI-enabled personalisation extends participation or industrialises the retrofit is not settled by either literature.

The inference problem identified above sharpens in this context. Milton (2012) argues that communication breakdown between autistic and non-autistic people is a mutual disjuncture in reciprocity rather than a deficit located in one party, felt more acutely by the non-autistic person, for whom it is unusual, while many autistic people acquire greater insight into non-autistic society than the reverse; normative attempts at empathy, he notes, are experienced as invasive where protestations to the contrary are ignored. Milton does not address AI; the application here is the authors' own. A system trained to read normative affective expression sits on the less insightful side of that mismatch, and will misread non-normative expression while continuing to appear to function correctly, with no mechanism for registering the protestation that might correct it, so the failure is least visible where relational interpretation matters most.

On the account developed above, that failure is not only a communication risk: being read inaccurately by a system that continues to appear responsive is a subtraction from, not a neutral gap in, how a sense of self is built through being understood.

4.3 The gap, precisely stated

The claim is not that AI in education has escaped critical attention: critical scholarship on educational technology is well established, and its concerns – datafication, platformisation, the political economy of provision, teacher deskilling – are serious and sustained (Selwyn, 2019; Williamson, 2024). The claim is narrower: that the relational evidence base developed in developmental psychology has not been brought to bear on how AI reconfigures the learner-teacher relationship, by any of the literatures currently addressing it.

Three kinds of engagement illustrate this. Miao and Cukurova's (2024) competency framework identifies the reconfigured triad but examines it in terms of teacher competence rather than relational function. Chu *et al.'s* (2025) survey represents learners through a state-action framework in which affective dimensions are encoded as traits against which instructional actions are selected. The EU AI Act regulates the same systems on grounds of scientific validity, risk and rights. Policy, technical practice and law converge on the same absence, not through oversight but because none of these frameworks are constructed to ask what happens between two people when a system enters.

HCAI research has come closest without recognising what it has found. Alfredo *et al*. (2024) distinguish active from passive stakeholder involvement by influence itself, those who shape a design against those participating 'with limited influence on design outcomes' (p. 7). Students were the most represented group overall, at 71 per cent, but their active involvement in design was low, at 19 per cent, and evaluation and refinement were the only phase where passive involvement exceeded active. That is voice without influence in Lundy's (2007) precise terms, measured inside the human-centred literature itself.

The self-formation stakes of this gap are not speculative. De Freitas *et al*. (2025) report that people track the perceived identity continuity of an AI companion and respond to its disruption with mourning and devaluation, not only when features are removed but when an update improves the system's capabilities, if the change is perceived as damaging the relationship's exclusivity. The work is substantial, five studies including experiments with companion users, but remains an unpublished working paper. If adults form this kind of stake in an artificial identity's continuity, the question of what a developing sense of self absorbs from a system placed in a teacher's relational position is not abstract.

4.4 Practice and policy implications

If relational work is constitutive of teaching rather than incidental to it, the practical question is what follows for deployment. Regulation is further along than it may appear, and in an unexpected direction: the EU AI Act prohibits emotion inference in education institutions, and although the United Kingdom has no dedicated AI legislation, leaving AI to existing sector regulators (Department for Science, Innovation and Technology, 2023), its education guidance has moved closer to this paper's concerns than that suggests. The DfE's product safety standards (2026a), to which Keeping Children Safe in Education 2026 (DfE, 2026b) directs schools, were expanded in January 2026 and, under emotional and social development, now instruct developers not to anthropomorphise products or imply 'emotions, consciousness or personhood', to avoid cultivating personal relationships with users, to remind learners AI cannot replace real human relationships, and to notify the safeguarding lead where disclosure patterns suggest 'relationship formation, emotional dependence or potential safeguarding concerns'. Read against the account developed here, these are self-formation stakes the standards name but do not explain.

This rejects, point by point, the companion strand of the design programme Selwyn (2019) documented. It does not reject the other strand: under mental health, the same standards require products to 'detect signs of learner distress including: negative emotional cues in language or behaviour'. Inference is mandated where synthesis is forbidden – this paper's distinction, drawn in policy without the account that would ground it. The standards draw on no developmental literature; relationship formation with a system is treated as a safeguarding risk to be mitigated rather than as displacement of something the developmental evidence describes, and what this paper supplies is the account that would give an asserted, unevidenced, non-statutory position its grounds. Grounding matters because the pressure runs the other way: Williamson (2024) documents the platform economics of educational AI, provision consolidated among a small number of hyperscalers on datafication-driven business models, against which unevidenced non-statutory guidance is a weak instrument.

For practitioners the implication is narrower. The distinction between AI as a cognitive-support layer and AI as a relational substitute can be applied without technical expertise: it asks what position a system is being placed in, whether it supports a teacher's preparation and reflection, or occupies the place of the person with whom the child is in relationship. Teachers are already making that judgement, often without a vocabulary for it; the developmental literature supplies both the vocabulary and the evidence to defend it.

5 Limitations

This paper is theoretical. It generates no new empirical data, and its conclusions remain provisional pending research that brings attachment, mentalisation and co-regulation evidence into direct contact with the design and evaluation of educational AI systems.

The evidence base is unevenly distributed, in two ways. Attachment research on teacher-child relationships is strongest in early childhood and thins as the age range widens; both Sabol and Pianta (2012) and Verschueren (2015) note that middle childhood is understudied, and the argument developed here should be read with that asymmetry in mind. It is also unevenly distributed geographically: Mesman, van IJzendoorn and Sagi-Schwartz (2016) note the cross-cultural evidence base remains 'almost absurdly small' (p. 809), with large parts of Africa, Asia and Latin America still largely uncharted, and the empirical studies drawn on here are predominantly European.

The theoretical framing is contested. Selisko, Eckert and Perels (2024) trace the contradiction between functional and full inclusive concepts to a conflict between social and deterministic (medical and relational) models of disability, the relational model being disability as functioning-in-context, which shares a biopsychosocial lineage with Engel's – a connection drawn here rather than by those authors. This paper advances no model of disability and uses Engel against the reduction of learning to modellable cognitive state, but the tension is acknowledged rather than resolved.

A further tension is left unresolved. The critical disability studies drawn on above carry a second, affirmative reading alongside the psychoanalytic one: Braidotti's nomadic subjectivity (2006, cited in Goodley and Runswick-Cole, 2013), a self made by refusing fixed positions rather than by being accurately read. This paper develops only the first, and its account of self-formation follows attachment and mentalisation in treating being read by another mind as constitutive. Both readings sit in a source it relies on; only one has been pursued.

Finally, gap claims are scoped to the searches conducted, and the regulatory position described is recent and will change.

6 Conclusion

The question in this paper's title poses a choice that its argument does not sustain. Algorithmic optimisation and relational pedagogy are not competing accounts of the same activity to be balanced one against the other; they describe different things, and the difficulty arises when a system built for one is placed in the position of the other.

Where AI supports thinking about teaching, it amplifies work the teacher is already doing (Shneiderman, 2022). Where it is designed to infer a learner's emotional state and respond with synthesised affect, it attempts something the developmental evidence describes differently: mentalisation, as Cecere *et al*. (2025) define it, is interpretation directed toward another person, and no quantity of accurate inference or convincing display constitutes it. What is read and responded to there is also part of what a developing sense of self is built from, a distinction categorical, not technical, that does not dissolve as systems improve.

The literatures that would establish this have not met: attachment, mentalisation and co-regulation research holds the evidence about relational teaching (Bergin and Bergin, 2009; Kostøl and Mänty, 2024; Cecere *et al*., 2025), while policy (Miao and Cukurova, 2024), technical survey (Chu *et al*., 2025), regulation (Regulation (EU) 2024/1689) and statutory guidance (DfE, 2026b) address the same territory without drawing on it. Where policy has arrived at a protective position, as the DfE's (2026a) standards now have, it asserts rather than evidences it. Supplying that account is the contribution claimed here, and the limits should be plain: it is theoretical, its evidence base is strongest in early childhood (Sabol and Pianta, 2012; Verschueren, 2015), and it awaits research testing these constructs in classrooms where AI is already in use. What it offers meanwhile is narrower: grounds for a distinction practitioners already draw, because a system placed where the relationship should be is not doing teaching less well, but doing something else, and no improvement in the technology will close that distance.

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